

day 2



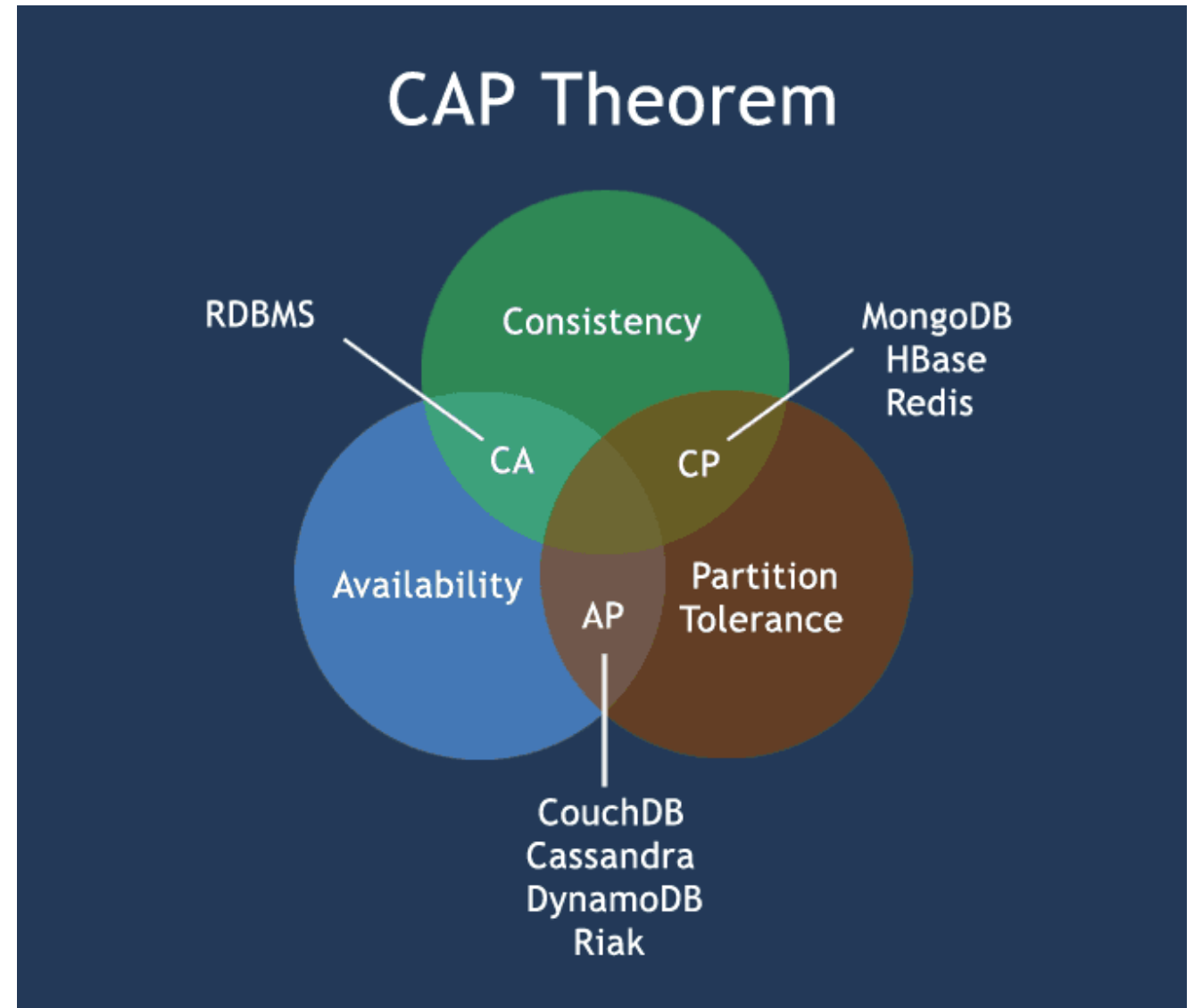
Fundamentals of Data Engineering

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CAP Theorem

- **Consistency** - Data is consistent after operation. After an update operation, all clients see the same data.
- **Availability** - System is always on (i.e. service guarantee), no downtime.
- **Partition Tolerance** - System continues to function even the communication among the servers is unreliable.



- ✓ **Key-value databases - e.g. redis, dynamodb, riak**

- Based on Amazon's Dynamo database.
- Keys are unique and values can be of any type i.e. JSON, BLOB, etc.
- Implemented as big distributed hash-table for fast searching.

Product ID	Type	Schema is defined per item		
1	Book ID	Odyssey	Homer	1871
2	Album ID	6 Partitas	Bach	
2	Album ID: Track ID	Partita No. 1		
3	Movie ID	The Kid	Drama, Comedy	Chaplin

RDBMS = 50000X30 29
1 state = count > 1000

- ✓ **Wide Column databases - e.g. hbase, cassandra, bigtable, ...** ^{google}

- Values of columns are stored contiguously.
- Better performance while accessing few columns & aggregations. ^{count, avg, sum, min, max}
- Good for data-warehousing, business intelligence, CRM, ...

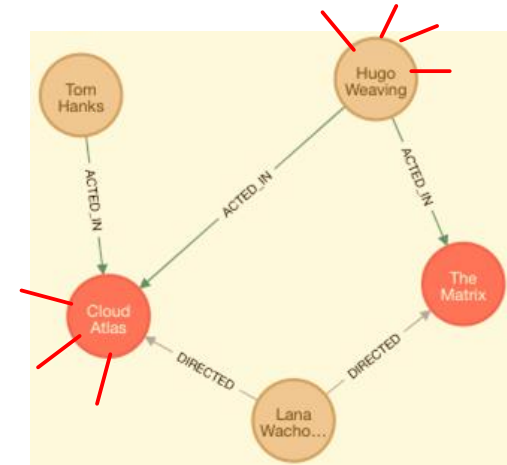
Row Key	Customer		Sales	
Customer Id	Name	City	Product	Amount
101	John White	Los Angeles, CA	Chairs	\$400.00
102	Jane Brown	Atlanta, GA	Lamps	\$200.00
103	Bill Green	Pittsburgh, PA	Desk	\$500.00
104	Jack Black	St. Louis, MO	Bed	\$1600.00



NoSQL Databases

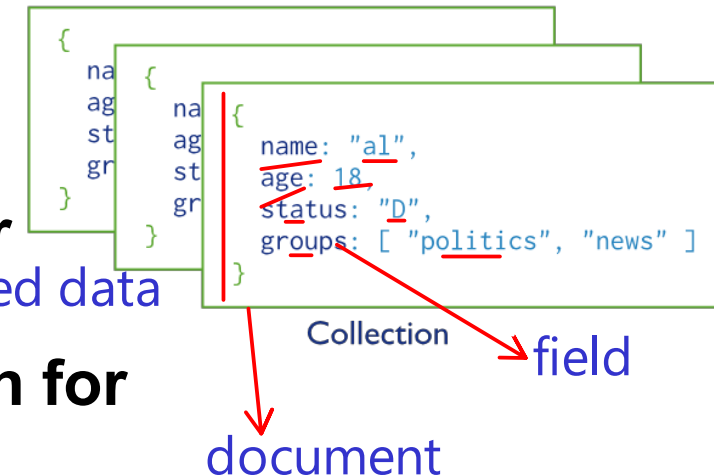
- Graph databases - e.g. Neo4J, Titan, ...

- Graph is collection of vertices and edges.
- Excellent performance, while dealing with all relations of an entity
(irrespective of size of data).



- Document oriented databases - e.g. MongoDb, CouchDb, ...

- Document contains data as key-value pair as JSON or XML.
- Document schema is flexible & are added in collection for processing.

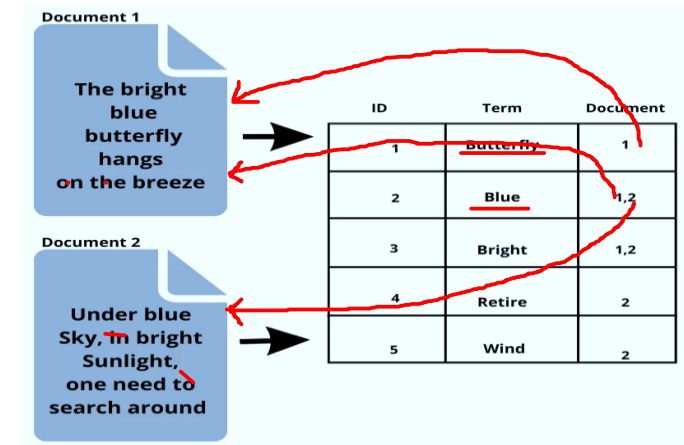


RDBMS	Doc-NoSQL
row	document
column	fields
table	collection

NoSQL Databases

google/yahoo search engine

- **Search databases – e.g. Elasticsearch, Solr, Lucene, ...**
 - For faster search – Text search, Log analysis.
 - Indexed, Exact/Fuzzy matches, Anomaly detection, Analytics.
- **Time series databases – e.g. Influx, Druid, ...**
 - Values organized by time like stock market, weather, ...
 - Optimized for retrieval, statistical processing, ...
 - Used for measurement data (weather, ...) and event-based data (accidents, ...)

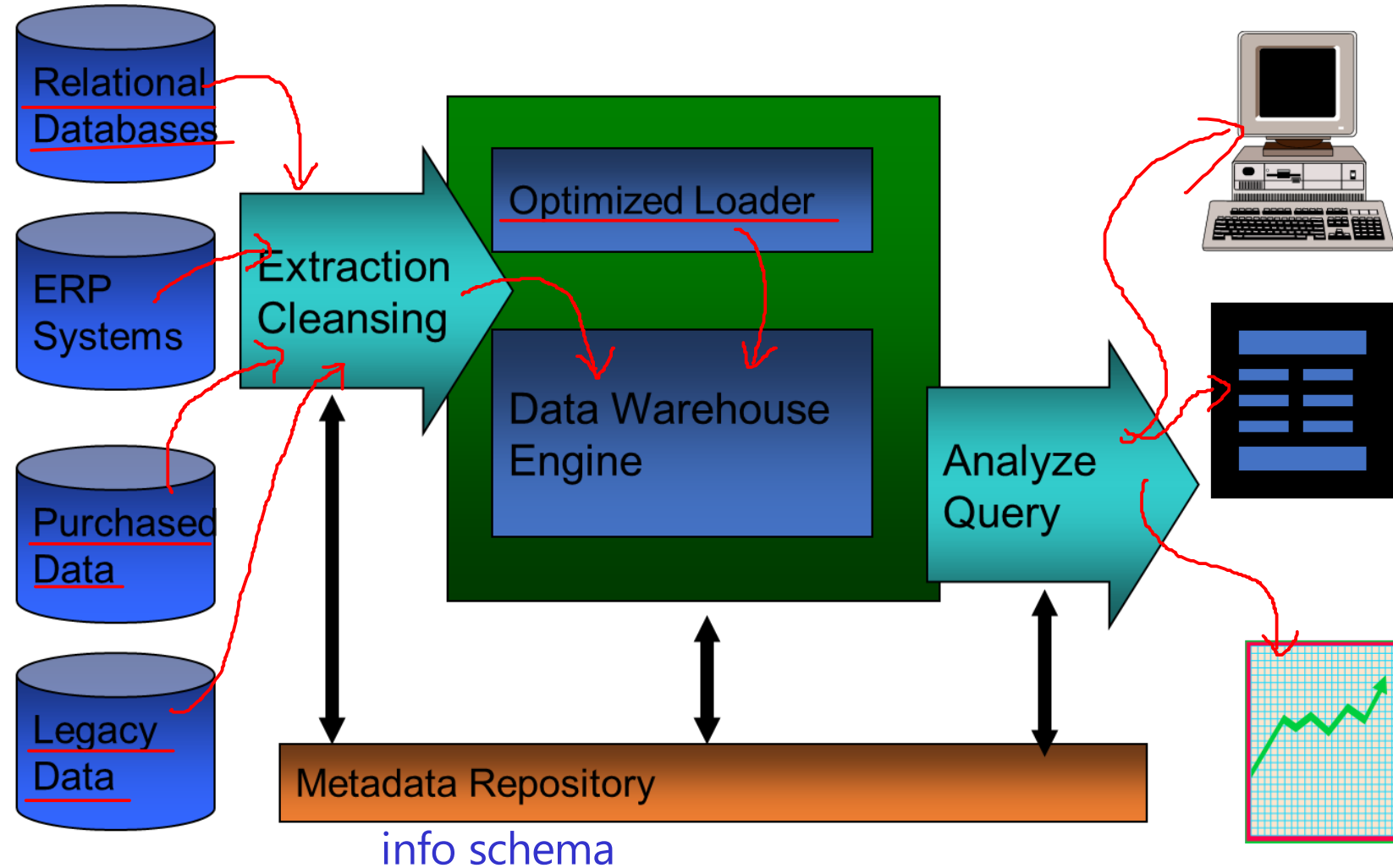


Date	Ozone ($\mu\text{g}/\text{m}^3$)	Temperature ($^{\circ}\text{C}$)	Relative humidity (%)	n deaths
1 Jan 2002	4.59	-0.2	75.7	199
2 Jan 2002	4.88	0.1	77.5	231
3 Jan 2002	4.71	0.9	81.3	210
4 Jan 2002	4.14	0.5	85.4	203
5 Jan 2002	2.01	4.3	93.5	224
6 Jan 2002	2.4	7.1	96.4	198



Data warehousing

- Data warehouse is a single, complete and consistent store of data obtained from a variety of different sources made available to end users in a what they can understand and use in a business context.
- Data warehousing is a process of transforming data into information and making it available to users in a timely enough manner to make a difference.

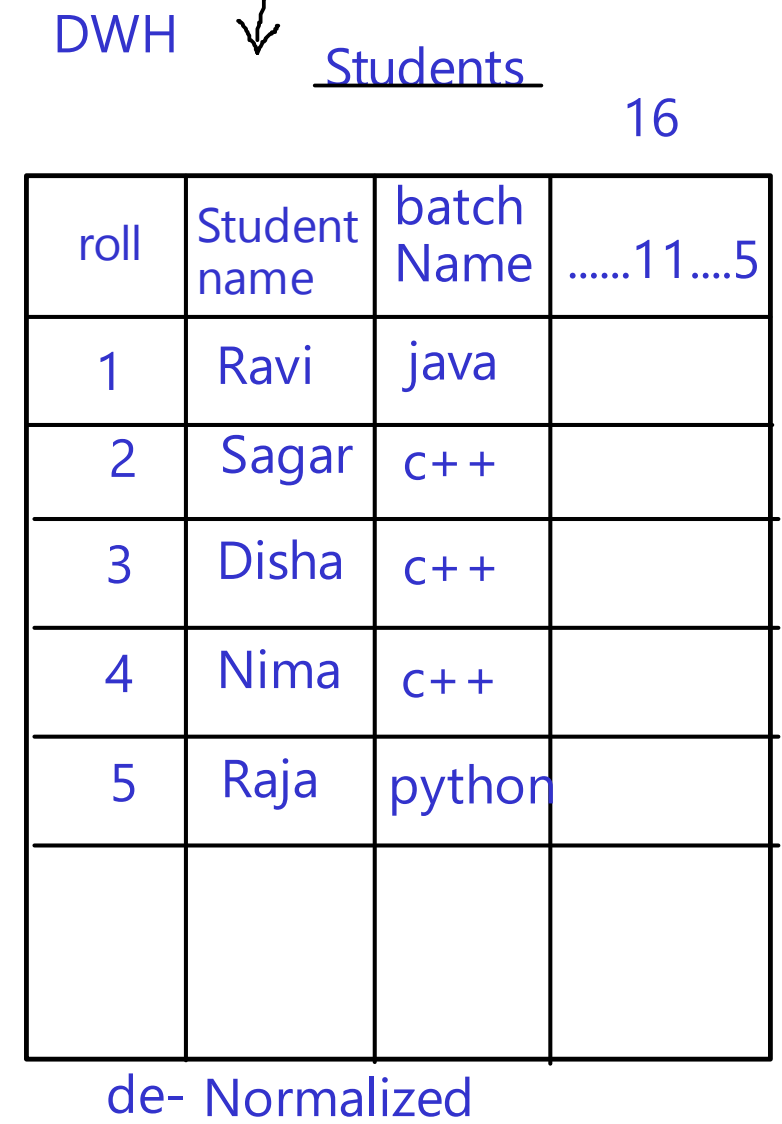
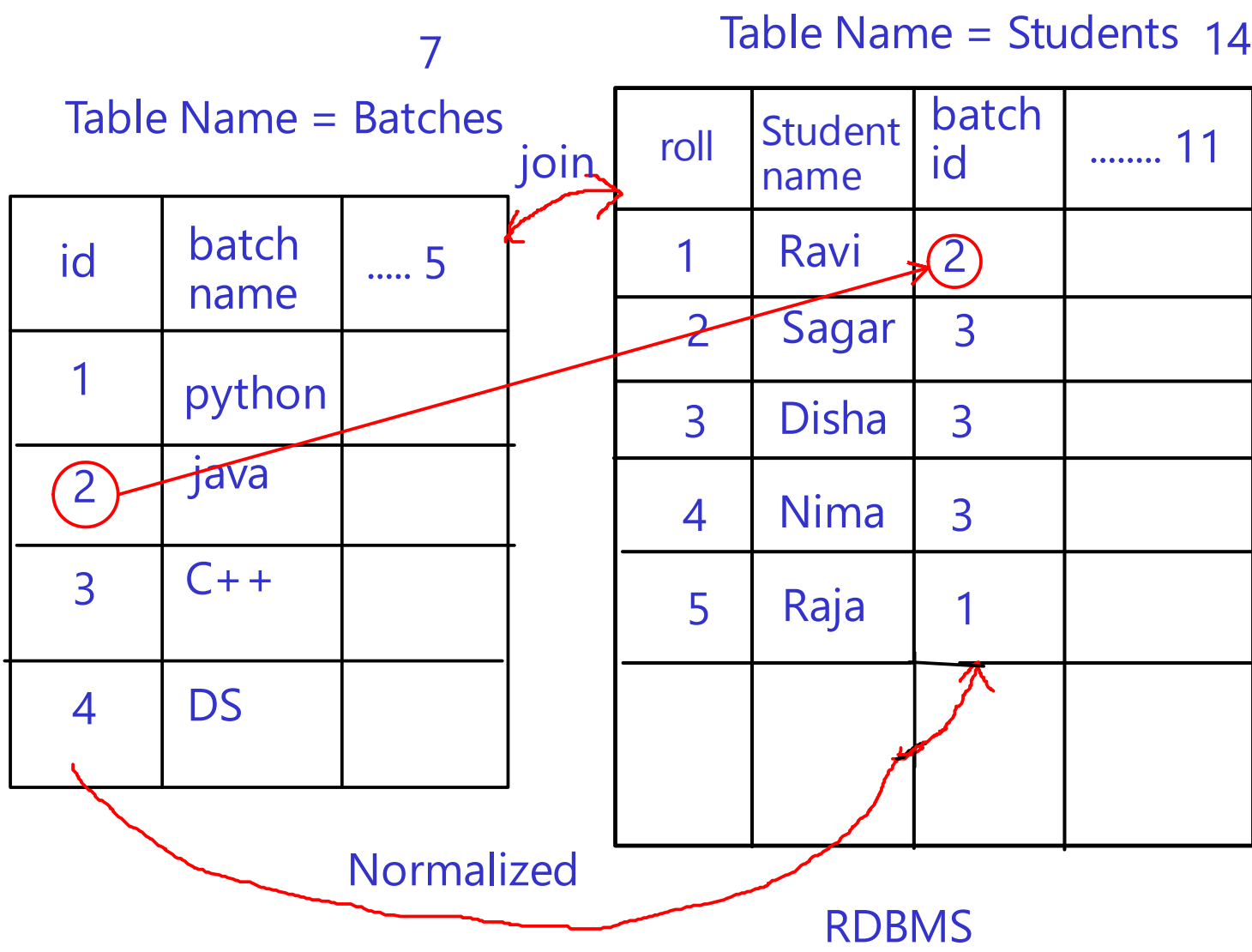


Extract – Transform → Load

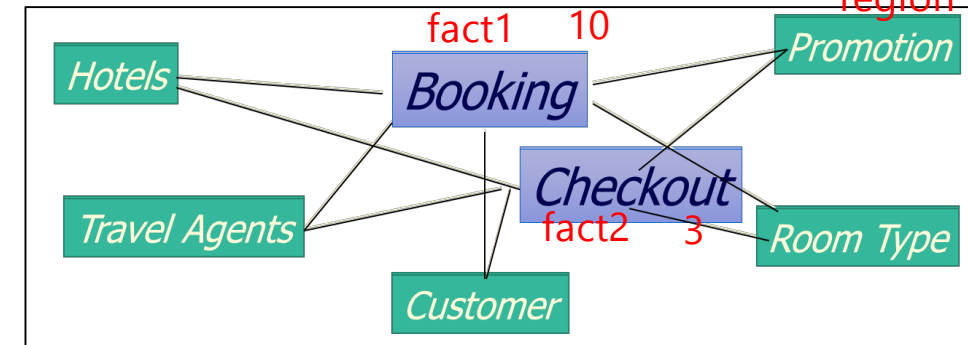
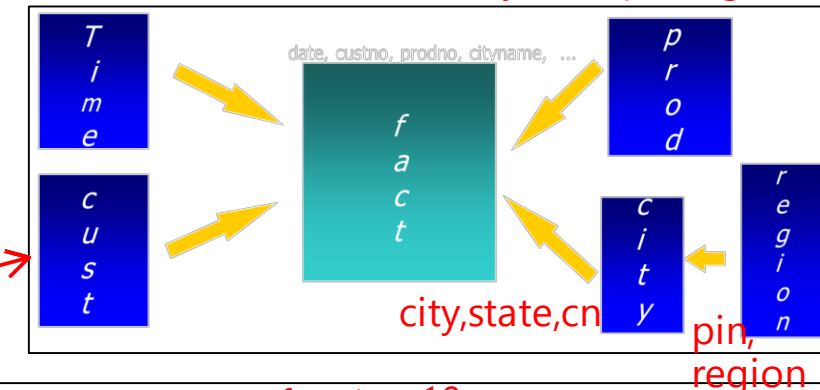
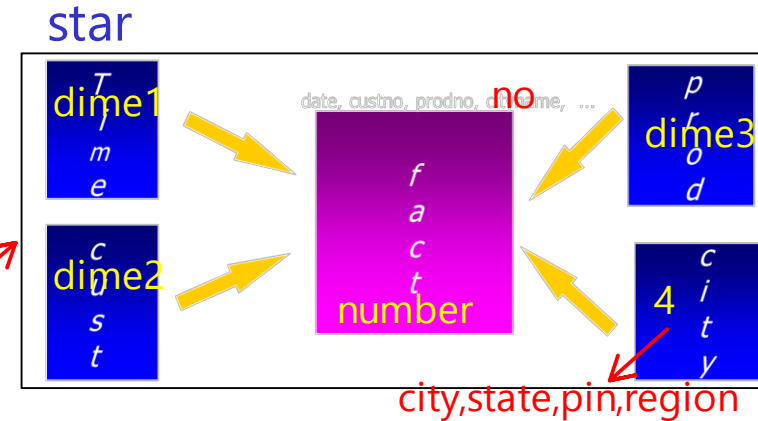
- Extracting: Extract data from sources into staging area
- Conditioning: Data types conversion to fit warehouse. mob ↗ ↘ str int
- House holding: Grouping similar data 1000/4 = 250
- Enrichment: Add relevant data from external sources pin->post
- Scoring: Computation of probability of an event
- Scrubbing: Data cleaning: find duplicate, missing data | | | |
- Merging: Merging data from various sources.
- De-normalize: Duplicate data to reduce joins.
- Loading: Load data in warehouse models like Star, Snowflake, Galaxy.
- Delta Updating: Incremental data uploading
- Partitioning: Dividing the data in logical parts to improve performance.



De-normalize :- Duplicate data to reduce joins.



- DWH schema is how data is stored in tables in warehouse for the efficient processing of the data.
- A fact table stores metrics, measurements, or facts about business processes. numbers
- Dimension tables are tables used to store data attributes or dimensions. number+ string
- Star schema: Single facts table and a few dimension tables (de-normalized) – Simple design.
- Snowflake schema: Single facts table and connected dimension/sub-dimension tables (normalized).
- Galaxy or Fact-Constellation schema: Multiple facts tables mapped to multiple dimension/sub-dimension tables.



OLTP (Database) vs OLAP (Data warehouse)

- **Online Transaction Processing**

- Modeled to run the business

- Detailed/Transactional normalized real-time data

- Transaction performance DML
DQL

- Read/Write operations

- Isolated data (Application specific)
 - Limited data (100 MB to 100 GB)

- **Online Analytical Processing**

- Modeled to analyze/optimize business

- Summarized/refined redundant snapshot data

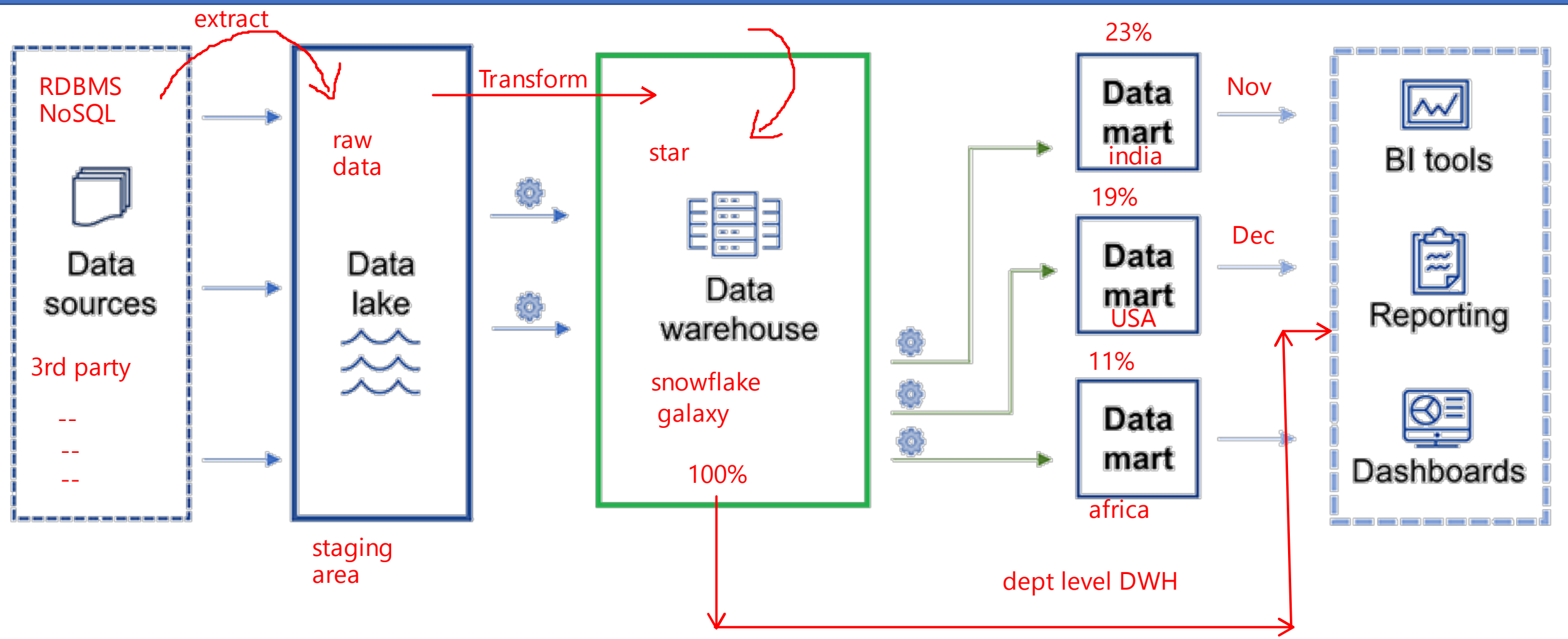
- Analytical query performance DQL

- Mostly Read operations

- Integrated data (from all sources) – Huge data (100 GB to Few TB)

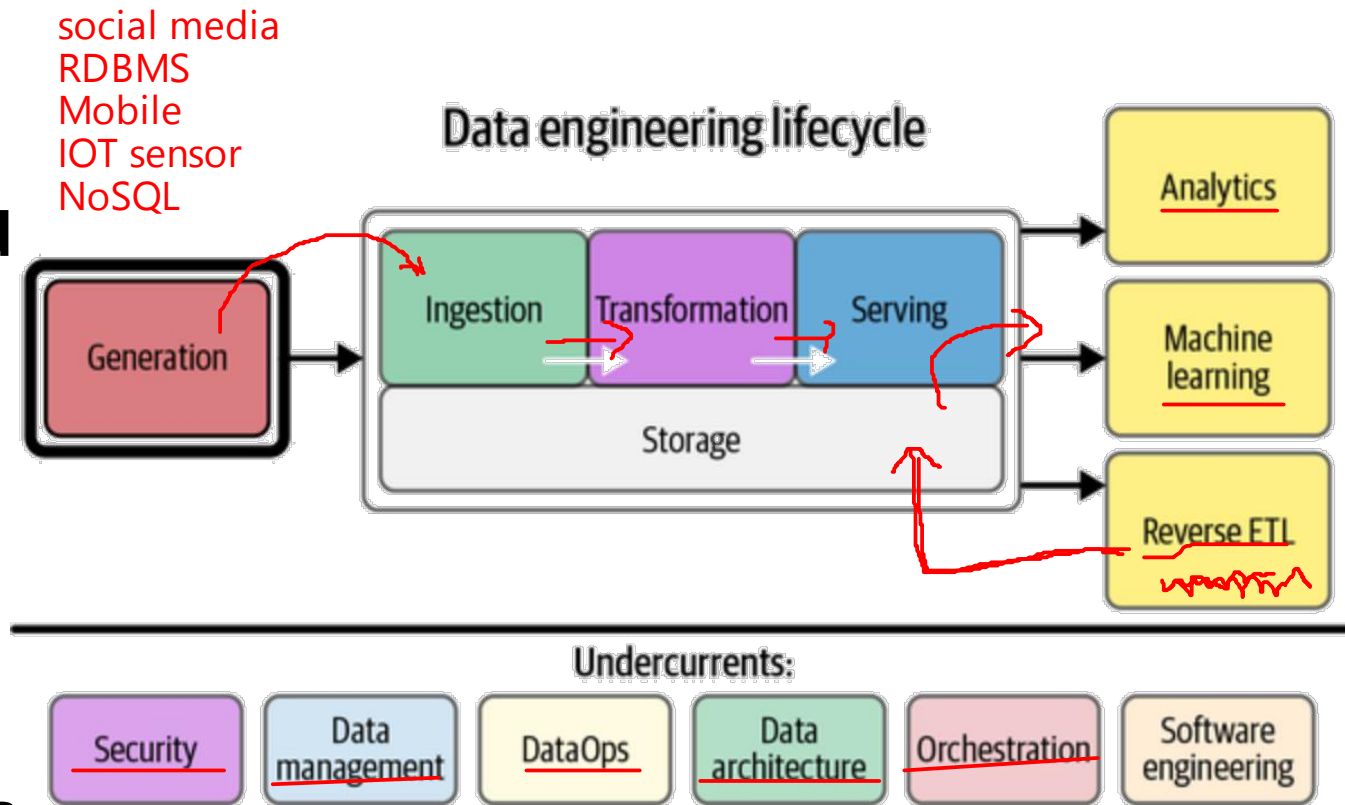


Data lake vs Data warehouse vs Data mart



Data engineering

- Data engineering is the development, implementation, and maintenance of systems and processes that take in raw data and produce high-quality, consistent information that supports downstream use cases, such as analysis and machine learning.
- Data engineer manages data engineering lifecycle, beginning with getting data from source systems & ending with serving data for use cases, such as analysis or machine learning.



Traditional ETL vs Hadoop ELT

old

new

- ETL stands for Extract, Transform and Load.
- The ETL process typically extracts data from the source/transactional systems, transforms it to fit the model of data-warehouse and finally loads it to the data warehouse.
- The transformation process involves cleansing, enriching and applying transformations to create desired output.
- Data is usually dumped to a staging area after extraction.

- 
- ELT stands for Extract, Load and Transform.
 - As opposed to loading just the transformed data in the target systems, the ELT process loads the entire data into the data lake. This results in faster load times.
 - The load process can also perform some basic validations and data cleansing rules.
 - The data is then transformed for analytical reporting as per demand.



Data storage

- Data storage is related to multiple stages in data engineering life cycle i.e. ingestion, transformation and serving.
- Storage needs to be selected based on read/write requirement, speed, durability, consistency, availability, scalability, fault tolerance, ... factors.
- Storage tradeoffs
 - Local storage vs Distributed storage
 - Strong consistency vs Eventual consistency
- Storage options are: File storage, Local disk storage, Network attached storage (NAS), Cloud file systems (S3/Blob), Block storage, RAID, Storage area network (SAN), Object storage, HDFS, Streaming storage.





Thank you!

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